

Project Initialization and Planning Phase

Date	23 June 2026
Team ID	XXX
Project Title	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report presents OptiCrop, a machine-learning-based agricultural decision-support system that recommends a suitable crop using soil nutrient and climatic parameters. The solution uses Nitrogen (N), Phosphorus (P), Potassium (K), temperature, humidity, pH and rainfall to provide a fast, data-driven crop recommendation through a simple web application.

Project Overview	
Objective	The primary objective of OptiCrop is to help users identify a suitable crop for given soil nutrient and climatic conditions using a trained machine-learning model.
Scope	The project focuses on crop recommendation using N, P, K, temperature, humidity, pH and rainfall. It provides a web-based interface for entering values and viewing the predicted crop and confidence result.
Problem Statement	
Description	Farmers and agricultural users may find it difficult to interpret multiple soil and climate parameters together when selecting a crop. Decisions based only on guesswork may not match actual field conditions.
Impact	An unsuitable crop choice can lead to poor productivity, inefficient use of resources, financial risk and loss of a cultivation season. Data-driven decision support can improve crop-planning confidence.
Proposed Solution	
Approach	OptiCrop uses a trained machine-learning model integrated with a Python Flask application. User inputs are validated and passed to the model, which returns a recommended crop and prediction confidence.
Key Features	ML-based Crop Recommendation N, P, K and Climate Input Form Input Validation Prediction Confidence Responsive Web Interface

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel/AMD Multi-Core Processor
Memory	RAM specifications	8 GB RAM or above
Storage	Disk space for data, models, and logs	Sufficient storage for dataset, model and project files
Software		
Frameworks	Python frameworks	Python, Flask
Libraries	Additional libraries	scikit-learn, pandas, NumPy, pickle
Development Environment	IDE	Visual Studio Code, HTML, CSS, JavaScript
Data		
Data	Source, size, format	Crop recommendation dataset (CSV) and trained model (crop_model.pkl)